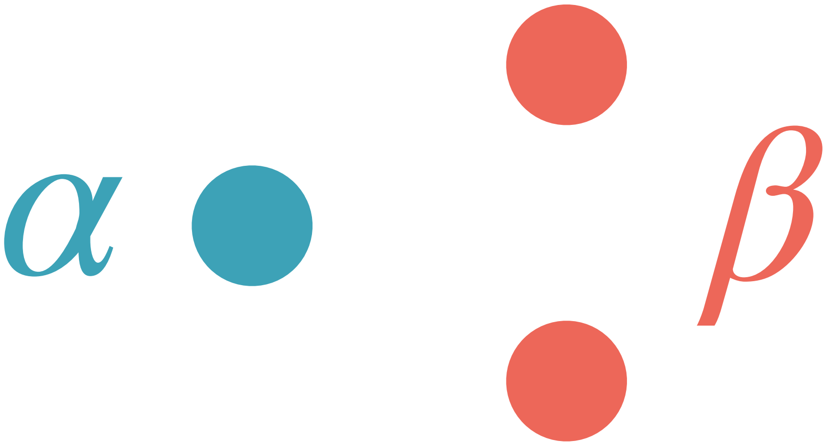
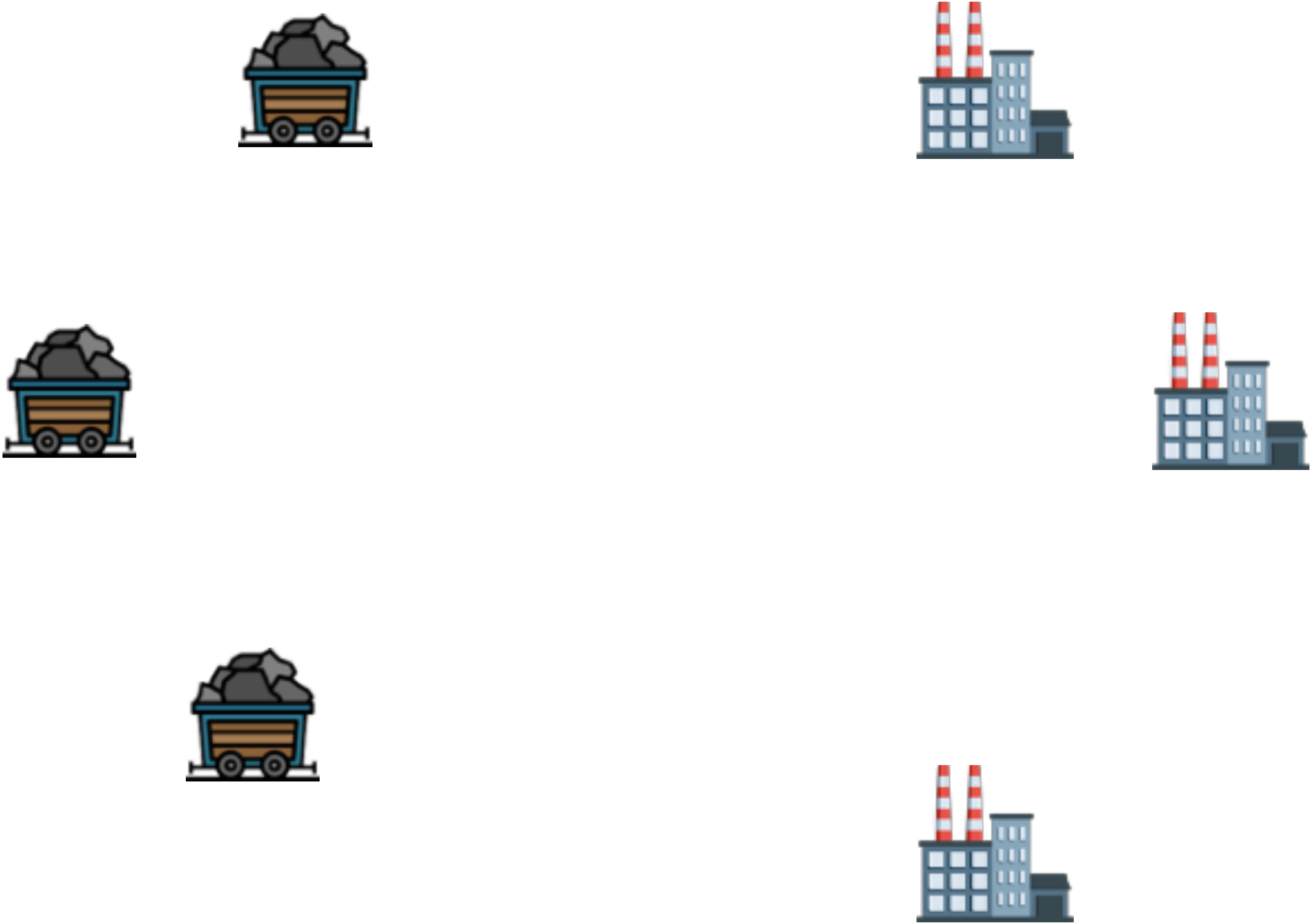
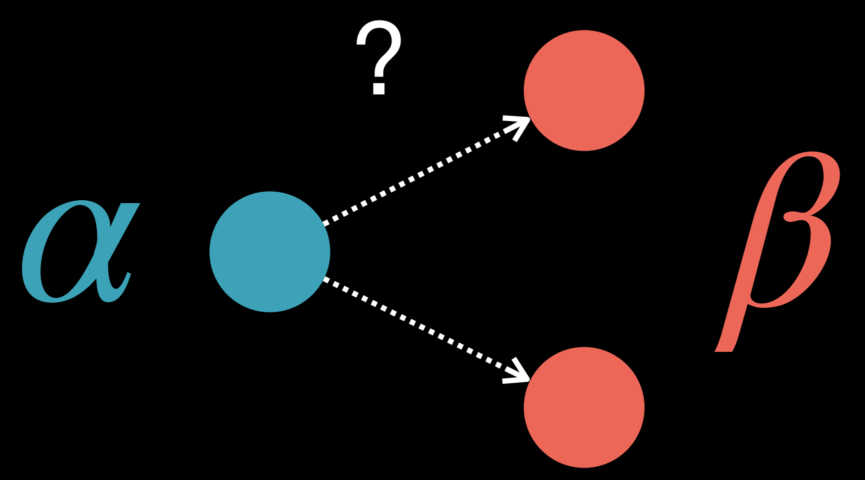


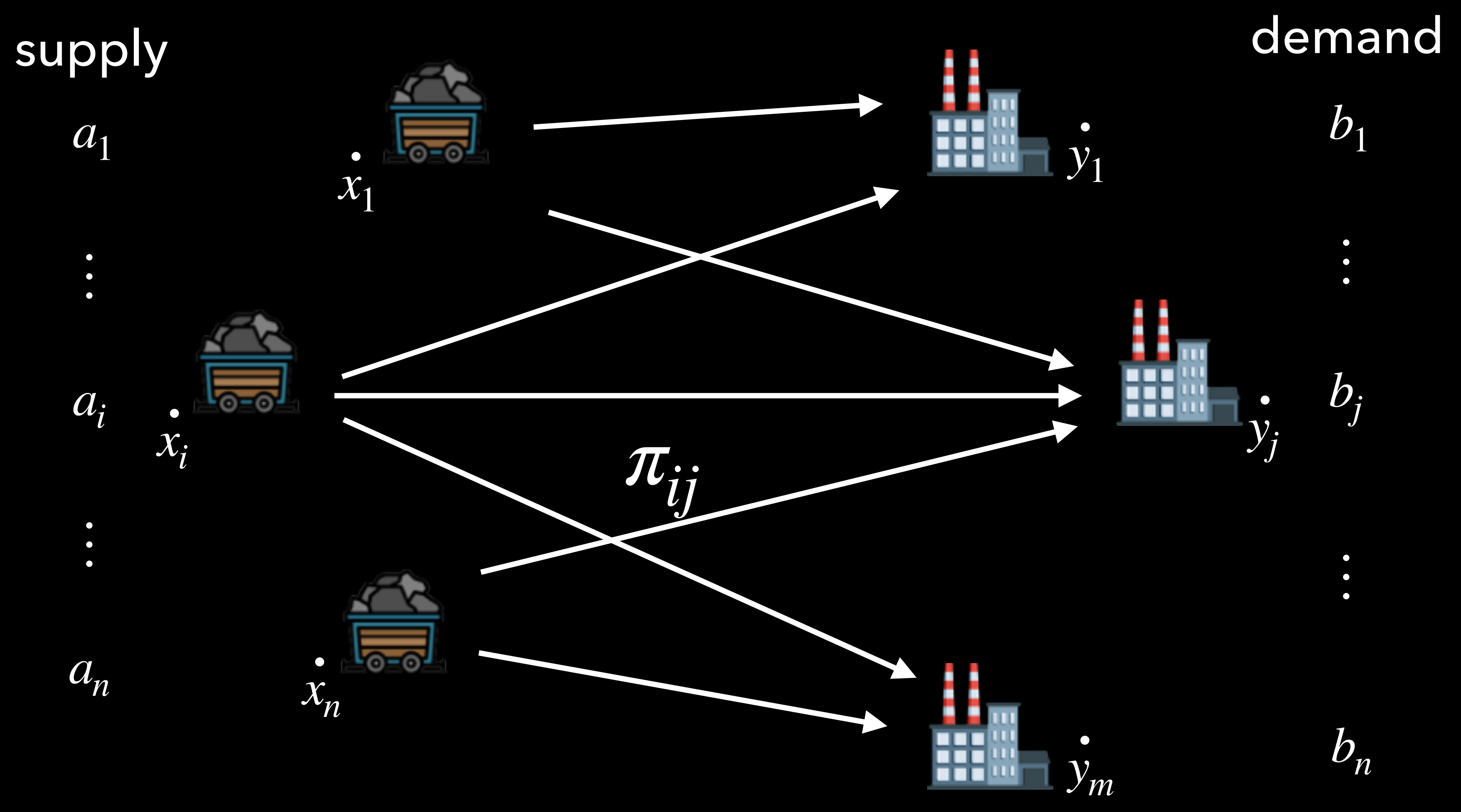
α
β



Splitting the Mass



$$a_i \neq b_j??$$



~~**Monge**: match i to j (~send all the mass in i to j)~~

Kantorovich: send π_{ij} mass from i to j

We call the matrix $[\pi_{ij}]_{i,j=1}^{n,m}$ a **coupling** between **a** and **b**

Couplings Formalize “Splitting”

Prob. measures:

$\alpha = \sum_1^n \mathbf{a}_i \delta_{x_i}, \mathbf{a} \in \Delta_n,$

$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$